THE ROTATED SPLIT TRIANGLE IS THE CONE TRIANGLE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3405-3421, following the statement at 3392-3403.

Editorial note. No omissions: both matrices are displayed and coincide; the identification of i and p with the natural maps is by construction.

Allowed standard prerequisites (editorial).

- Associated triangle of a termwise split sequence. definition-split-ses and the associated triangle construction (same chapter, cited at ~3400). Supplies delta and the matrix form of the differential.

Lemma 0.1. *Let $\mathcal{A}$ be an additive category. Let $(\alpha : A^\bullet \to B^\bullet, \beta : B^\bullet \to C^\bullet, s^n, \pi^n)$ be a termwise split sequence of complexes. Let $(A^\bullet, B^\bullet, C^\bullet, \alpha, \beta, \delta)$ be the associated triangle. Then the triangle $(C^\bullet[-1], A^\bullet, B^\bullet, \delta[-1], \alpha, \beta)$ is isomorphic to the triangle $(C^\bullet[-1], A^\bullet, C(\delta[-1])^\bullet, \delta[-1], i, p)$.*

Proof. We write $B^n = A^n \oplus C^n$ and we identify α^n and β^n with the natural inclusion and projection maps. By construction of δ we have

$$d_B^n = \begin{pmatrix} d_A^n & \delta^n \\ 0 & d_C^n \end{pmatrix}$$

On the other hand the cone of $\delta[-1] : C^\bullet[-1] \to A^\bullet$ is given as $C(\delta[-1])^n = A^n \oplus C^n$ with differential identical with the matrix above! Whence the lemma. $\square$

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.